USN. | | | | | | | | | | [24MTCSE101]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY-2025

DESIGN AND ANALYSIS OF ALGORITHMS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	**Marks**
1.	**a.**	Define Big oh, Small OH, Big Omega, Small Omega.	CO1	L2	**10**

	b.	Analyze the space complexity of the following pseudo code.
		1) Fun (INTEGER n)
		{
		IF (n>= 0)
		{
		Fun(n-1)
		PRINT (Design and Analysis of Algorithm)
		}
		2) Analyze the time complexity of the following code.
		Fun ()
		{
		INTEGER i=1, s=1
		For (i=1,s=1; s<=n; i=i+1,s=s+i)
		PRINT (Design and Analysis of Algorithm)
		}
			CO1	L3	**10**

2.	**a.**	Explain Merge sort algorithm with an example.	CO3	L3	**10**
	b.	Analyze the time complexity and space complexity of Merge sort algorithm.	CO2	L3	**10**
3.	**a.**	Explain with an example that how to detect cycle using disjoined set structure.	CO3	L2	**10**
	b.	Design an efficient algorithm for the following scenario. Given are N ropes of different lengths, the task is to connect these ropes into one rope with minimum cost, such that the cost to connect two ropes is equal to the sum of their lengths. Example: ***Input:*** *arr[] = {4,3,2,6} , N = 4* ***Output:*** *29* ***Explanation:*** 1. *First, connect ropes of lengths 2 and 3. Now we have three ropes of lengths 4, 6, and 5.* 2. *Now connect ropes of lengths 4 and 5. Now we have two*	CO3	L2	**10**

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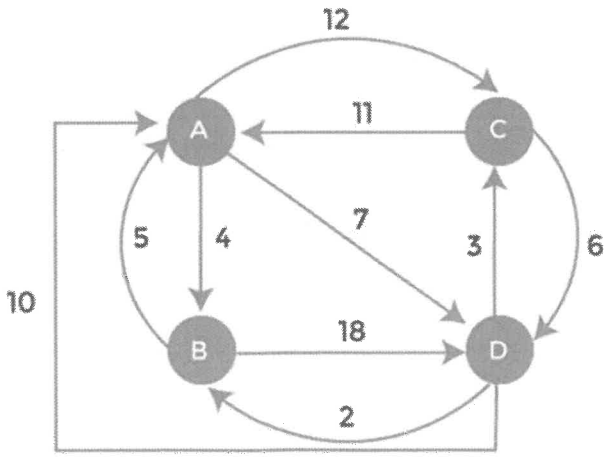
		ropes of lengths 6 and 9. 3. *Finally connect the two ropes and all ropes have connected.*			
		OR			
4.	**a.**	Explain Heap sort algorithm with an example.	CO3	L2	**10**
	b.	How to use heap data structure in shortest path algorithm?	CO3	L2	**10**
5.	**a.**	You are given a knapsack with a capacity of W = 50. There are 4 items with the following weights and values: Item | Weight (W) | Value (V) 1 | 10 | 60 2 | 20 | 100 3 | 30 | 120 4 | 40 | 200 Determine the maximum value that can be carried in the knapsack and the items selected.	CO4	L3	**10**
	b.	Explain the difference between divide and conquer strategy and dynamic programming with necessary example.	CO4	L3	**10**
		OR			
6.	**a.**	Give an efficient algorithm for the following scenario. Given an array array of non-negative integers and a value sum, the task is to check if there is a subset of the given array whose sum is equal to the given sum.	CO5	L3	**10**
	b.	Analyse the role of Branch and Bound in solving the Travelling Salesman Problem (TSP) through a step-by-step illustration for the given problem  	CO5	L3	**10**
7.	**a.**	Define NP-complete, NP – hard problems with example.	CO6	L4	**10**
	b.	Identify the Vertex Cover Problem in detail, focusing on its classification as an NP-complete problem and its applications. Explain how approximation algorithms are used to solve it in practical scenarios.	CO5	L3	**10**
		OR			

8.	**a.**	Prove formally that Clique is an NP – complete problem.	CO6	L4	**10**
	b.	Prove that Vertex cover is an NP – complete problem	CO6	L4	**10**